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Development of Fault Tolerant Software for Embedded Realtime Systems

Project:

Fault Tolerance Jump Counter for Skipping Rope Using T-Watch

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Objective of the project

Develop a smartwatch application to:

- Count skipping rope jumps
- Measure workout duration
- Provide basic workout control (start, stop, reset)
- Demonstrate fault tolerant behaviour
- Minimize required user interaction during workout execution



System Overview

- User starts workout manually
- Watch reads IMU acceleration data continuously
- Jump detection logic counts valid jump events
- Timer runs in parallel with jump detection
- Fault tolerant logic monitors:
 - Jump inactivity
 - Sensor plausibility



Fault Tolerant Pattern 1

Minimize Human Intervention

- **Pattern:** Minimize Human Intervention (Hanmer)
- **Intent:** System should react automatically without requiring user input.
- **Implementation:**
 - If no jump is detected for 15 seconds:
 - Workout stops automatically
 - Timer freezes
 - System enters a safe and idle state



Why It Is Fault Tolerance

- Prevents invalid jump detections caused by sensor noise
- Avoids incorrect jump counts and workout duration
- Ensures consistent internal system state

Result:

- System remains reliable despite IMU uncertainty
- Erroneous data is contained and corrected early



Fault Tolerant Pattern 2

Correcting Audits

- **Pattern:** Correcting Audits (Hanmer)
- **Intent:** Detect and correct erroneous data before it propagates through the system.
- **Implementation:**
 - IMU sensor data is continuously checked for plausibility
 - Invalid or unrealistic acceleration peaks are discarded
 - Minimum time gap between consecutive jumps is enforced
 - System state remains consistent despite noisy sensor data



Why It Is Fault Tolerance

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Demo

Video 1



Video 2





Limitations

- **Single axis measurement:** Only Z axis acceleration used; tilt or lateral motion may cause missed jumps or false detections.
- **Loose watch wearing:** Extra wrist movement can generate false acceleration peaks.
- **False positives:** Sudden hand/arm movements (swings, claps) may be counted as jumps.
- **No orientation compensation:** Watch rotation reduces Z axis accuracy.
- **Single sensor system:** No redundancy; system relies solely on built in IMU.



Conclusion

- A fault tolerant jump counter smartwatch application was implemented
- Two architectural fault tolerance patterns (Hanmer) were applied:
 - Minimize Human Intervention
 - Correcting Audits
- System behaves robustly under real world conditions with noisy sensor data
- Fault tolerance improves reliability without increasing user effort

Thank you !